

VOCALIS: A Lightweight Neural Adapter for Synthetic-to-Natural Speaker Embedding Adaptation

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Abstract—Recent advances in zero-shot speech synthesis and pretrained speaker representation models have enabled synthetic speech to be increasingly integrated into modern speech processing systems. However, speaker embeddings extracted from synthetic speech often exhibit a significant distribution mismatch with embeddings derived from natural recordings, reducing their compatibility with downstream speaker-conditioned speech processing systems. This paper presents *VOCALIS*, a lightweight neural adapter that learns a mapping from synthetic speaker embeddings to the natural embedding space by improving alignment with corresponding natural speaker embeddings. The proposed approach employs a residual multilayer perceptron trained using paired natural–synthetic speaker embeddings extracted from the VCTK corpus. Unlike conventional approaches, only the adapter network is optimized while the pretrained speaker encoder remains fixed, enabling efficient adaptation with minimal computational overhead. Experimental evaluation on speaker-disjoint training, validation, and test partitions shows a substantial improvement in cosine similarity from 0.0081 to 0.3239 between synthetic and natural speaker embeddings. Additional experiments demonstrate consistent performance across multiple random seeds, successful generalization to an unseen speech synthesis system (gTTS), and improved adaptation quality with increasing amounts of training data. These results suggest that lightweight embedding adaptation is an effective and computationally efficient strategy for reducing the synthetic-to-natural speaker embedding gap.

Index Terms—Speaker Embedding, Domain Adaptation, Zero-Shot Speech Synthesis, Speech Representation Learning, Synthetic Speech, Deep Learning

I. INTRODUCTION

Recent advances in neural speech technologies have transformed applications such as text-to-speech (TTS) synthesis, speaker verification, voice conversion, and personalized conversational systems. Central to many of these applications is the use of speaker representations, typically encoded as fixed-dimensional embeddings that capture speaker-specific characteristics while remaining largely independent of linguistic content. These representations have become an essential component of modern speech processing pipelines, enabling reliable speaker conditioning, identity preservation, and speaker-aware generation across a wide range of downstream tasks.

Alongside these developments, synthetic speech has emerged as an increasingly valuable resource for speech technology. Modern neural TTS systems are capable of generating

highly natural speech, making synthetic data attractive for data augmentation, low-resource language development, privacy-preserving applications, and large-scale dataset construction. As the quality and availability of synthetic speech continue to improve, many contemporary speech systems increasingly incorporate synthetic audio during training and evaluation. Consequently, representations extracted from synthetic speech are expected to coexist with representations obtained from natural human recordings within the same processing pipelines.

Despite this growing reliance on synthetic speech, an important assumption remains largely unexplored. Most speaker representation models are developed and optimized using natural speech corpora. When these pretrained encoders are directly applied to synthetic speech, however, the resulting embeddings are not guaranteed to occupy the same representation space as those extracted from natural recordings. Differences introduced by speech synthesis, including spectral artifacts, prosodic variations, and generation-specific acoustic characteristics, can introduce a measurable distribution mismatch between the two domains. Such discrepancies may reduce the compatibility of synthetic speaker embeddings with downstream systems that have been designed primarily around natural speech representations.

Existing research has largely focused on improving speech generation quality, speaker similarity, and zero-shot voice conversion, while advances in speaker representation learning have concentrated on developing increasingly discriminative embedding models such as GE2E, x-vectors, and ECAPA-TDNN. More generally, domain adaptation techniques have demonstrated considerable success in mitigating distribution shifts across diverse machine learning problems. Nevertheless, comparatively little attention has been devoted to explicitly adapting speaker embeddings extracted from synthetic speech to better align with the natural speech embedding space. As synthetic speech becomes increasingly prevalent in modern speech processing pipelines, reducing this embedding-space discrepancy represents an important yet underexplored research problem.

To address this challenge, we propose **VOCALIS**, a lightweight neural domain adaptation method that learns to transform speaker embeddings extracted from synthetic speech

into representations that more closely resemble those derived from natural speech. Rather than modifying existing speaker encoders or retraining downstream speech models, VOCALIS operates entirely within the embedding space, allowing it to be incorporated into existing speech processing pipelines with minimal computational overhead. The proposed model employs a residual multilayer perceptron trained using paired synthetic–natural embedding supervision together with a composite objective that simultaneously encourages directional similarity, positional alignment, and transformation regularization.

Unlike conventional domain adaptation approaches designed for generic feature alignment, VOCALIS specifically targets speaker representation adaptation. The proposed methodology is supported by a reproducible experimental pipeline incorporating speaker-disjoint data partitioning, paired embedding construction, rigorous train–validation–test separation, and multiple evaluation protocols to ensure fair and unbiased assessment.

The effectiveness of the proposed method is evaluated using paired natural and synthetic speech derived from the VCTK corpus. The experimental study includes comparisons with classical linear adaptation methods, evaluation on previously unseen speakers, cross-synthesis generalization using an unseen text-to-speech system, multi-seed reproducibility analysis, and data scaling experiments. Across these evaluations, VOCALIS consistently reduces the discrepancy between synthetic and natural speaker embeddings while maintaining a compact architecture suitable for practical deployment.

The primary contributions of this work are summarized as follows:

- We formulate synthetic-to-natural speaker embedding adaptation as a dedicated embedding-space domain adaptation problem motivated by the increasing integration of synthetic speech into modern speech processing systems.
- We propose VOCALIS, a lightweight residual neural adaptation method that learns an explicit mapping between synthetic and natural speaker embedding spaces without modifying pretrained speaker encoders or downstream speech models.
- We establish a reproducible experimental protocol incorporating speaker-disjoint evaluation, paired embedding supervision, classical linear adaptation baselines, cross-synthesis generalization, reproducibility analysis, and data scaling experiments.
- We demonstrate that the proposed method consistently improves the alignment between synthetic and natural speaker embeddings while maintaining a computationally efficient architecture that can be readily integrated into existing speech processing pipelines.

II. RELATED WORK

A. Speaker Representation Learning

Speaker representations have become a fundamental component of modern speech processing systems, providing compact fixed-dimensional embeddings that characterize speaker

identity independently of linguistic content. Recent advances in deep speaker representation learning have enabled highly discriminative embeddings that support applications such as speaker verification, speaker diarization, voice conversion, and speaker-conditioned speech synthesis. Comparative studies of ECAPA-TDNN and x-vector representations demonstrate that modern embedding architectures provide improved robustness and generalization for zero-shot multi-speaker speech synthesis [1]. Furthermore, large-scale self-supervised learning models such as WavLM have shown that pretrained speech representations can effectively transfer across numerous downstream speech processing tasks [2].

Despite these advances, existing work primarily focuses on improving the discriminative capability of speaker representations themselves. Comparatively little attention has been devoted to investigating whether speaker embeddings extracted from synthetic speech remain consistent with those extracted from natural speech when processed by pretrained speaker encoders.

B. Zero-Shot Speech Synthesis

Recent years have witnessed rapid progress in zero-shot speech synthesis through increasingly powerful neural architectures capable of generating high-quality speech for previously unseen speakers. Early work such as YourTTS demonstrated that multilingual zero-shot speech synthesis and voice conversion could be achieved within a unified framework using speaker conditioning [3]. More recent systems including NaturalSpeech 3, VALL-E 2, StyleTTS-ZS, and DS-TTS further improved speech naturalness, speaker similarity, and robustness through neural codec language models, diffusion-based generation, style modeling, and adaptive speaker conditioning [4]–[7].

Several studies have also investigated lightweight adaptation mechanisms for improving zero-shot speech synthesis. Adapter-based architectures and self-supervised conditioning methods demonstrate that parameter-efficient modules can substantially improve synthesis quality without requiring complete retraining of large speech generation models [8]–[10]. These approaches highlight the effectiveness of lightweight adaptation strategies within modern speech synthesis systems.

Nevertheless, the primary objective of existing zero-shot TTS systems remains high-quality speech generation. The compatibility of speaker embeddings extracted from synthetic speech with those derived from natural human speech has received comparatively little attention.

C. Domain Adaptation for Speech Representations

Domain adaptation seeks to reduce distribution shifts between different data domains while preserving task-relevant information. Classical methods such as Deep CORAL align feature distributions by minimizing covariance discrepancies between source and target domains [11]. Adversarial approaches such as Domain-Adversarial Neural Networks (DANN) instead learn domain-invariant representations through adversarial optimization [12]. These techniques have demonstrated strong

performance across computer vision, natural language processing, and general representation learning.

Within speech processing, recent work has also explored cross-lingual voice transfer and robust speaker conditioning across different synthesis settings [13]. However, these approaches primarily focus on improving speech generation or voice transfer quality rather than explicitly adapting speaker embeddings extracted from synthetic speech into the natural embedding space.

D. Research Gap and Motivation

The existing literature demonstrates substantial progress in speaker representation learning, zero-shot speech synthesis, and general domain adaptation. However, comparatively little research has explicitly addressed the discrepancy between speaker embeddings extracted from synthetic speech and those extracted from natural recordings while preserving compatibility with existing pretrained speaker encoders and downstream speech systems.

Rather than introducing a new speech synthesis architecture or proposing a new speaker encoder, the proposed VOCALIS framework addresses this gap by operating entirely within the embedding space. VOCALIS learns a lightweight neural mapping from synthetic speaker embeddings to their corresponding natural representations using paired supervision, enabling synthetic embeddings to become more consistent with natural embedding distributions without modifying existing speech generation or speaker representation models.

III. METHODOLOGY

A. Problem Definition

The objective of VOCALIS is to reduce the discrepancy between speaker embeddings extracted from synthetic speech and those extracted from natural speech while preserving compatibility with existing pretrained speaker encoders and downstream speech processing systems. Let $\mathbf{e}_s \in \mathbb{R}^d$ denote the speaker embedding extracted from a synthetic speech utterance and $\mathbf{e}_n \in \mathbb{R}^d$ denote the corresponding embedding extracted from the paired natural recording of the same speaker and transcript. VOCALIS learns a lightweight transformation

$$f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^d \quad (1)$$

such that

$$f_\theta(\mathbf{e}_s) \approx \mathbf{e}_n, \quad (2)$$

where θ represents the parameters of the proposed adaptation network. Unlike conventional approaches that retrain the speaker encoder or modify the speech synthesis model, VOCALIS keeps the pretrained speaker encoder fixed and performs adaptation entirely within the embedding space. Consequently, the framework can be incorporated into existing speech processing pipelines without altering the underlying speech generation or speaker representation models.

B. Overview of the VOCALIS Framework

The overall framework is illustrated in Fig. 2. Natural speech recordings are obtained from the VCTK corpus, while paired synthetic speech is generated offline using a text-to-speech system with identical transcripts. Both recordings are processed using the same pretrained speaker encoder to produce paired speaker embeddings.

During training, the synthetic embedding serves as the input to the proposed adapter network and the corresponding natural embedding acts as the supervision target. The adapter learns a direct mapping between the two embedding domains using paired supervision. During inference, only a synthetic speech recording is required. The extracted synthetic embedding is passed through the trained adapter to obtain an adapted embedding that can be directly supplied to downstream speaker-conditioned applications.

C. Dataset Construction

VOCALIS employs the VCTK speech corpus as the source of natural speech recordings. To ensure unbiased evaluation, speakers are partitioned into mutually exclusive training, validation, and testing subsets using a speaker-disjoint protocol. Consequently, no speaker appears in more than one partition throughout the entire experimental pipeline.

For every selected utterance, synthetic speech is generated offline using Piper while preserving the original transcript. This produces paired natural and synthetic recordings containing identical linguistic content but different acoustic realizations. Metadata describing speaker identity, transcript, dataset partition, and audio paths is maintained throughout the pipeline, enabling deterministic pairing between natural and synthetic samples.

Prior to model training, a verification stage validates the generated dataset by checking speaker-disjoint partitions, paired sample consistency, embedding availability, and metadata integrity. This verification step ensures that all subsequent experiments operate on a consistent and reproducible dataset.

D. Speaker Embedding Extraction

Speaker embeddings are extracted independently from both natural and synthetic speech using a pretrained speaker encoder. The implementation supports multiple embedding backbones, including ECAPA-TDNN and Resemblyzer, allowing VOCALIS to remain independent of any specific speaker representation model. Rather than specifying the embedding dimensionality manually, the implementation automatically infers the embedding size from the selected encoder, enabling seamless compatibility across different embedding models.

To improve computational efficiency, embeddings are extracted only once during preprocessing and cached to disk. Consequently, the training stage operates exclusively on paired embedding vectors rather than repeatedly processing raw audio, substantially reducing computational cost while ensuring deterministic model inputs throughout all experiments.

E. Residual Adapter Network

VOCALIS employs a lightweight residual adapter network that transforms synthetic speaker embeddings toward the natural speaker embedding space. The adapter consists of multiple fully connected layers interleaved with Layer Normalization, Rectified Linear Unit (ReLU) activations, and dropout regularization. A residual connection allows the transformed representation to preserve speaker-specific information from the original embedding while learning only the correction required to reduce the synthetic-to-natural domain discrepancy.

Unlike conventional end-to-end adaptation methods, the pretrained speaker encoder remains fixed throughout training. Only the parameters of the adapter network are optimized. Finally, the adapted embedding is L2-normalized before being returned by the network, ensuring consistent embedding magnitude and improved compatibility with cosine similarity, which serves as the primary evaluation metric.

F. Training Objective

The adapter is trained using paired synthetic–natural speaker embeddings and a composite objective designed to encourage both directional similarity and positional alignment within the embedding space. The overall objective is defined as

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{cos}} + \lambda_2 \mathcal{L}_{\text{mse}} + \lambda_3 \mathcal{L}_{\text{reg}}, \quad (3)$$

where λ_1 , λ_2 , and λ_3 control the relative contribution of each component.

The cosine similarity loss encourages the adapted embedding to align with the target natural embedding,

$$\mathcal{L}_{\text{cos}} = 1 - \frac{\hat{\mathbf{e}}_s^T \mathbf{e}_n}{\|\hat{\mathbf{e}}_s\| \|\mathbf{e}_n\|}, \quad (4)$$

where $\hat{\mathbf{e}}_s$ denotes the adapted synthetic embedding.

To additionally encourage positional alignment in the embedding space, mean squared error is minimized,

$$\mathcal{L}_{\text{mse}} = \|\hat{\mathbf{e}}_s - \mathbf{e}_n\|_2^2. \quad (5)$$

A regularization term further constrains the learned transformation to avoid unnecessarily large deviations from the original synthetic embedding, improving optimization stability while preserving speaker identity.

G. Training Procedure

Only the adapter parameters are optimized during training, while the pretrained speaker encoder remains fixed. Optimization is performed exclusively on the training speaker partition, whereas model selection is based on validation performance. The held-out test speakers are never accessed during optimization, thereby preventing train–test leakage.

Training employs mini-batch optimization with the Adam optimizer. Validation loss is monitored after each training epoch, and the checkpoint achieving the best validation performance is retained for all subsequent experiments. This checkpoint is subsequently used for held-out evaluation, cross-synthesis generalization, reproducibility analysis, residual-weight ablation, and training-data scaling experiments.

H. Inference Pipeline

During inference, VOCALIS requires only a synthetic speech recording. The recording is processed using the pretrained speaker encoder to obtain a synthetic speaker embedding, which is then passed through the trained adapter network to produce an adapted embedding. The output of VOCALIS is therefore an adapted speaker embedding rather than synthesized audio. Since neither the speaker encoder nor the downstream speech processing system is modified, the proposed framework can be integrated into existing speech processing pipelines as a lightweight intermediate adaptation module.

IV. EXPERIMENTAL SETUP

A. Dataset

Experiments are conducted using the VCTK speech corpus [14], which contains multi-speaker English speech recordings from speakers with diverse regional accents. To evaluate the ability of the proposed method to generalize to previously unseen speakers, a speaker-disjoint evaluation protocol is adopted throughout all experiments. The dataset is partitioned into independent training, validation, and test subsets containing 28, 6, and 6 speakers, respectively. No speaker appears in more than one partition, thereby preventing speaker leakage between training and evaluation.

For each selected speaker, up to 45 utterances are retained to provide a balanced representation across the dataset while maintaining a consistent number of paired samples across speakers. Every natural recording is paired with a synthetic counterpart generated from the same transcript, producing paired natural–synthetic speech samples that differ only in their acoustic realization. Metadata describing speaker identity, transcript, dataset partition, and audio paths is stored throughout the pipeline to ensure deterministic pairing and reproducibility. The complete dataset preparation workflow is illustrated in Fig. 1.

B. Synthetic Speech Generation

Synthetic speech is generated offline using the Piper text-to-speech system. Each transcript from the natural VCTK corpus is synthesized independently while preserving the original linguistic content. This procedure produces paired natural and synthetic utterances that share identical text but exhibit different acoustic characteristics due to the speech synthesis process.

To evaluate the robustness of the proposed adaptation framework beyond the synthesis model observed during training, an additional cross-synthesis experiment is performed using Google Text-to-Speech (gTTS). Importantly, the adapter is trained exclusively on Piper-generated speech and is never exposed to gTTS during optimization. Consequently, the cross-synthesis evaluation measures the ability of VOCALIS to generalize across previously unseen speech synthesis systems.

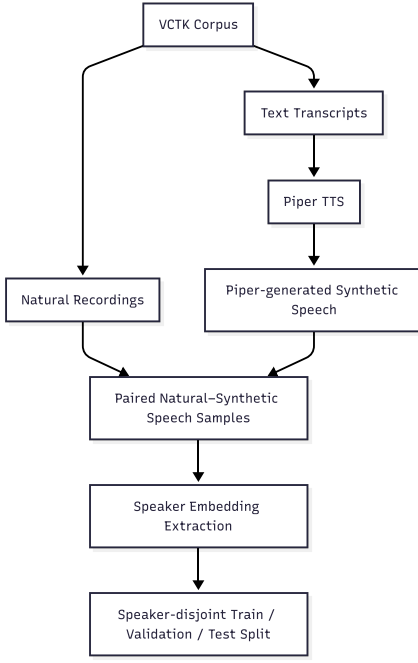


Fig. 1. Pipeline for constructing paired natural-synthetic speech samples from the VCTK corpus.

C. Speaker Embedding Extraction

Speaker embeddings are extracted using a pretrained speaker encoder. The implementation supports both ECAPA-TDNN and Resemblyzer backbones, allowing the framework to remain independent of any specific speaker representation architecture. Rather than manually specifying the embedding dimensionality, the implementation automatically infers the embedding size from the selected encoder, enabling seamless compatibility across different embedding models.

To improve computational efficiency, speaker embeddings are extracted once during preprocessing and cached to disk. Model training therefore operates directly on paired embedding vectors instead of repeatedly processing raw audio, substantially reducing computational cost while ensuring deterministic inputs throughout all experiments.

D. Training Configuration

Only the proposed adapter network is optimized during training, while the pretrained speaker encoder remains fixed. Model parameters are optimized using the Adam optimizer with mini-batch gradient descent. Validation loss is monitored after every training epoch, and the checkpoint achieving the lowest validation loss is retained for all subsequent experiments. The held-out test partition is never accessed during optimization, ensuring a strict separation between training and evaluation.

E. Baseline Methods

To determine whether the nonlinear adapter provides meaningful improvements beyond simple distribution alignment techniques, VOCALIS is compared against multiple baseline

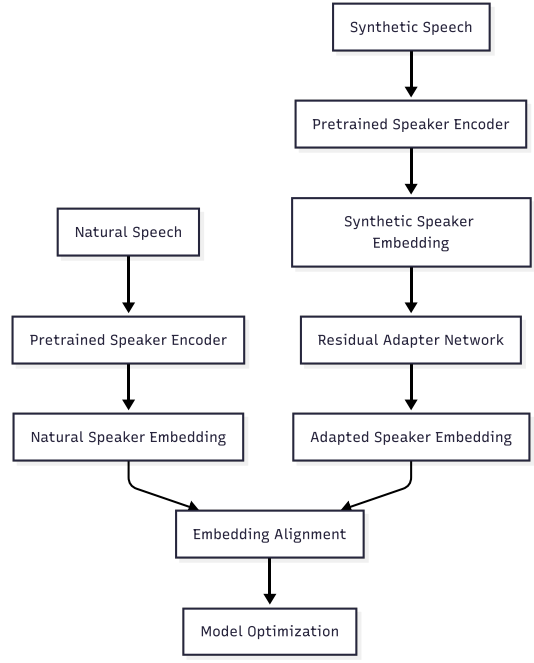


Fig. 2. Overview of the proposed VOCALIS framework.

TABLE I
EXPERIMENTAL CONFIGURATION USED THROUGHOUT THE EVALUATION.

Parameter	Value
Dataset	VCTK Corpus
Training Speakers	28
Validation Speakers	6
Test Speakers	6
Utterances per Speaker	Up to 45
Synthetic Speech Engine	Piper
Cross-TTS Evaluation	gTTS
Primary Speaker Encoder	ECAPA-TDNN
Alternative Encoder	Resemblyzer
Embedding Storage	Cached Embeddings
Adapter Architecture	Residual Adapter Network
Normalization	LayerNorm + L2 Norm
Activation Function	ReLU
Optimizer	Adam
Training Strategy	Best Validation Checkpoint
Evaluation Metric	Cosine Similarity

approaches. The evaluated baselines include identity mapping (no adaptation), global mean-shift alignment, and Deep CORAL-inspired covariance alignment. All methods are evaluated under the same speaker-disjoint protocol using identical training, validation, and test partitions.

F. Evaluation Metrics

Performance is primarily evaluated using cosine similarity between adapted synthetic embeddings and their corresponding natural speaker embeddings. Cosine similarity directly measures the consistency of speaker identity within the embedding space and therefore serves as the principal evaluation

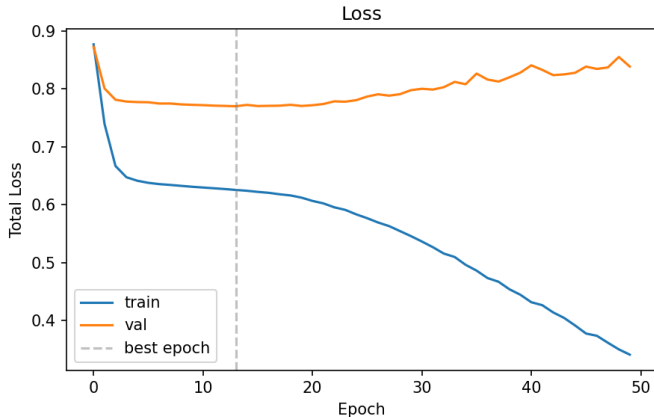


Fig. 3. Training and validation loss during adapter optimization.

TABLE II
AGGREGATE TEST-SET PERFORMANCE: NO ADAPTATION VS. VOCALIS.

Metric	No Adaptation	VOCALIS	Δ
Cosine Similarity $\uparrow$	0.0081	0.3239	+0.3158
Euclidean Distance $\downarrow$	1.4077	1.1601	-0.2476
MSE $\downarrow$	0.0103	0.0070	-0.0033

metric throughout the paper. Aggregate statistics and per-speaker performance are both reported.

G. Experimental Protocol

The proposed evaluation protocol is designed to assess not only adaptation performance but also robustness and reproducibility. Evaluation includes held-out speaker testing, cross-synthesis generalization using gTTS, multi-seed reproducibility analysis, residual-weight ablation, and training-data scaling experiments.

V. RESULTS

A. Domain Gap Reduction on Held-Out Speakers

Table II reports aggregate performance on the held-out test partition (6 speakers, 270 utterances), comparing the raw synthetic embedding against the corresponding natural embedding (no adaptation) with the output of the trained VOCALIS adapter. Figure 3 illustrates the training and validation loss during adapter optimization, with the selected checkpoint corresponding to the minimum validation loss. Without adaptation, synthetic and natural embeddings are nearly uncorrelated, with a mean cosine similarity of only 0.0081. Following adaptation, mean cosine similarity increases to 0.3239, an absolute improvement of 0.3158. Consistent improvements are observed for Euclidean distance and mean squared error.

Table III presents the corresponding per-speaker breakdown. All six held-out test speakers exhibit a positive improvement in cosine similarity following adaptation, with no speaker showing degraded performance. The magnitude of improvement varies across speakers (0.2328 to 0.4305), which we

TABLE III
PER-SPEAKER COSINE SIMILARITY ON HELD-OUT TEST SPEAKERS.

Speaker	Baseline	VOCALIS	Δ	Std
p253	-0.0247	0.4058	+0.4305	0.0532
p257	0.0245	0.4276	+0.4032	0.0795
p281	-0.0260	0.2178	+0.2438	0.0491
p317	0.0120	0.2449	+0.2328	0.0579
p340	0.0122	0.3504	+0.3382	0.0726
p351	0.0504	0.2970	+0.2465	0.0601

TABLE IV
COMPARISON OF ADAPTATION METHODS ON HELD-OUT TEST SPEAKERS.

Method	Mean Cosine Similarity
No Adaptation	0.0081
Mean-Shift	0.2051
Deep CORAL [11]	0.1747
VOCALIS (proposed)	0.3239

attribute to natural variation in how distinctly each speaker’s synthetic and natural acoustic characteristics differ under the fixed speaker encoder.

To assess whether this consistent improvement is statistically meaningful given the number of held-out speakers, we apply a one-sided sign test at the speaker level (the appropriate unit of independence, since utterances from the same speaker are correlated). With all 6 of 6 test speakers showing improvement, the sign test yields $p = (1/2)^6 = 0.0156$, below the conventional 0.05 significance threshold. Although the number of independent test speakers is limited, the consistent improvement observed across all held-out speakers yields statistical significance under the one-sided sign test.

B. Comparison with Baseline Adaptation Methods

To determine whether the nonlinear residual adapter provides meaningful benefit over simpler distribution alignment techniques, we compare VOCALIS against two classical linear baselines: a global mean-shift (translating the synthetic embedding distribution to match the natural embedding mean) and Deep CORAL [11] (aligning full covariance structure). Both baselines are fit exclusively on the training partition and evaluated on the same held-out test speakers, using identical L2-normalized cosine-similarity scoring.

Both linear baselines meaningfully reduce the domain gap relative to no adaptation, confirming that the synthetic-to-natural discrepancy has exploitable linear structure. However, VOCALIS outperforms the stronger of the two linear baselines (mean-shift, 0.2051) by 0.1188, and outperforms Deep CORAL by 0.1492, suggesting that the nonlinear transformation learned by the residual adapter captures structure beyond what a closed-form linear alignment can recover at this data scale. We note that the simpler mean-shift baseline outperforms full covariance-based CORAL alignment; we discuss this observation further in Section VI.

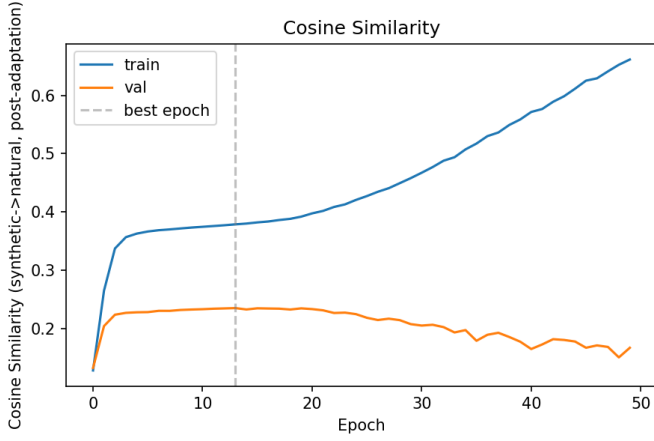


Fig. 4. Training and validation cosine similarity during adapter optimization.

TABLE V
TEST COSINE SIMILARITY ACROSS 5 RANDOM SEEDS.

Seed	Test Cosine Similarity	Improvement
42	0.3223	+0.3143
123	0.3261	+0.3180
7	0.3234	+0.3153
2024	0.3249	+0.3169
99	0.3235	+0.3154
Mean $\pm$ Std	0.3240 $\pm$ 0.0015	+0.3160 $\pm$ 0.0015

C. Reproducibility Across Random Seeds

To verify that the reported improvement reflects a stable learned effect rather than a favorable initialization, we retrain the adapter from scratch across 5 random seeds, holding all other configuration fixed, and evaluate each resulting model on the same held-out test partition. Figure 4 shows the evolution of cosine similarity during optimization, indicating stable convergence and supporting the selection of the best validation checkpoint for evaluation.

Across all 5 seeds, test cosine similarity ranges from 0.3223 to 0.3261, a spread of only 0.0038, corresponding to a relative variability of approximately 0.5% of the mean. This indicates that the reported improvement is a stable, reproducible property of the trained model on this dataset rather than an artifact of a particular random initialization.

D. Cross-Synthesis Generalization

The adapter is trained exclusively on Piper-synthesized speech. To evaluate whether the learned transformation generalizes to a synthesis system never observed during training, we additionally synthesize the same test-speaker transcripts using gTTS and evaluate the trained adapter on the resulting embeddings without any retraining or fine-tuning.

Despite never encountering gTTS-derived embeddings during training, the adapter improves mean cosine similarity from 0.0507 to 0.3165 on gTTS audio, comparable to its in-distribution performance on Piper. All 6 test speakers show

TABLE VI
CROSS-SYNTHESIS GENERALIZATION: PIPER (TRAINED ON) VS. gTTS (UNSEEN).

Synthesis System	Baseline Cosine	Adapted Cosine
Piper (in-distribution)	0.0081	0.3239
gTTS (unseen)	0.0507	0.3165

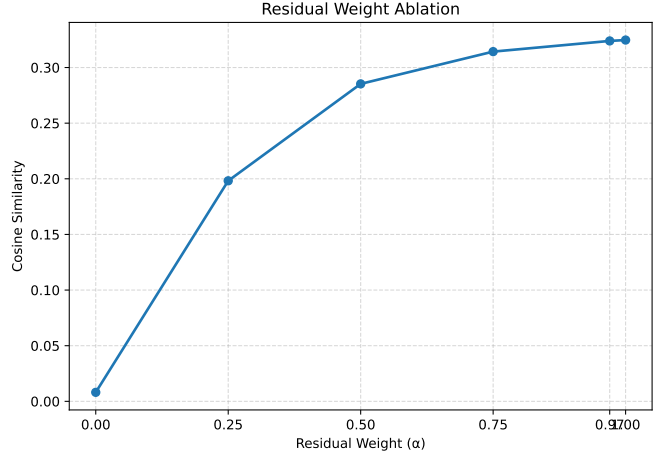


Fig. 5. Effect of the residual blending coefficient on cosine similarity.

positive improvement under gTTS as well (range: 0.1595 to 0.3644). The near-zero baseline observed for both synthesis systems (0.0081 for Piper, 0.0507 for gTTS) further confirms that the domain gap is a general property of synthetic speech under this speaker encoder, rather than an artifact specific to a single TTS system.

E. Residual Weight Analysis

The adapter architecture includes a learnable scalar α (Section III) controlling the blend between the original synthetic embedding and the learned MLP correction, with $\alpha = 0$ corresponding to an identity mapping. Figure 5 illustrates the effect of varying the residual blending coefficient on adaptation performance, while Table VII reports the corresponding quantitative results.

At $\alpha = 0$, test cosine similarity (0.0081) is identical to the no-adaptation baseline reported in Table II, confirming that the residual connection correctly reduces to an identity mapping as designed. The learned value of α after training converged to 0.9696, indicating that the trained model relies almost entirely on the MLP transformation rather than the residual pass-through component.

F. Effect of Training Data Scale

To assess whether the observed improvement reflects a genuine learned structure that benefits from additional data, rather than an artifact of one particular training set size, we retrain the adapter using progressively smaller subsets of the 28 training speakers, evaluating each resulting model on the same fixed 6-speaker test partition. Figure 6 illustrates

TABLE VII
TEST COSINE SIMILARITY AS A FUNCTION OF RESIDUAL WEIGHT α .

Residual Weight (α)	Test Cosine Similarity
0.00 (identity mapping)	0.0081
0.25	0.1982
0.50	0.2853
0.75	0.3143
0.97 (as trained)	0.3239
1.00	0.3247

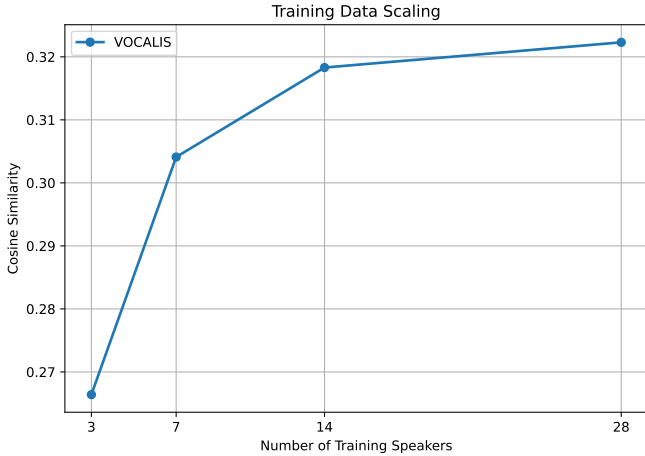


Fig. 6. Effect of training data size on adaptation performance.

the relationship between training data size and adaptation performance, while Table VIII summarizes the corresponding quantitative results.

Test cosine similarity increases monotonically with the number of training speakers, with diminishing returns as the training set grows. This trend suggests that VOCALIS learns a generalizable mapping between synthetic and natural embedding spaces, and suggests that further performance gains may be achievable by scaling training data beyond the 28 speakers used in this study.

VI. DISCUSSION

The experimental results indicate that the discrepancy between synthetic and natural speaker embeddings constitutes a meaningful domain adaptation problem rather than merely an artifact of a particular speaker encoder or speech synthesis system. Across all evaluation settings, the proposed VOCALIS framework consistently improves the alignment between synthetic and natural speaker embeddings while maintaining a lightweight architecture that operates entirely in the embedding space. These findings suggest that explicit embedding adaptation can serve as an effective intermediate processing stage for speech systems that incorporate synthetic speech without requiring modifications to existing speaker encoders or downstream models.

A notable observation is that even simple linear transformations substantially reduce the synthetic-to-natural embedding

TABLE VIII
TEST COSINE SIMILARITY AS A FUNCTION OF TRAINING SPEAKER COUNT.

Train Speakers	Train Pairs	Test Cosine Similarity
3	135	0.2664
7	315	0.3041
14	630	0.3183
28	1260	0.3223

gap. Both the mean-shift baseline and Deep CORAL improve cosine similarity relative to the unadapted embeddings, indicating that a significant portion of the domain discrepancy can be explained through global distribution mismatch. Interestingly, the global mean-shift transformation outperforms covariance alignment in the present experiments. One possible explanation is the relatively small amount of paired training data available for estimating full covariance statistics, making the simpler mean translation more robust under the current data regime. Nevertheless, the proposed nonlinear residual adapter consistently achieves the strongest performance, suggesting that the relationship between synthetic and natural speaker embeddings cannot be fully captured using linear transformations alone.

The cross-synthesis experiments further indicate that the learned transformation generalizes beyond the text-to-speech system used during training. Although VOCALIS is optimized exclusively using Piper-generated speech, it maintains comparable performance on gTTS-generated utterances without retraining or fine-tuning. This result suggests that the learned mapping captures characteristics of the broader synthetic-to-natural embedding discrepancy rather than merely memorizing artifacts specific to a single speech synthesis model. Such robustness is particularly important as modern speech processing pipelines increasingly integrate outputs from multiple neural speech generation systems.

The multi-seed experiments indicate that the proposed optimization procedure is highly stable. Minimal variation across independent random initializations suggests that the observed improvements are reproducible rather than dependent on favorable optimization trajectories. Similarly, the residual-weight analysis shows that the learned blending coefficient converged close to one, implying that the network relies predominantly on the learned nonlinear transformation while retaining the flexibility of residual learning during optimization. This behavior supports the design choice of incorporating a residual connection without constraining the model to preserve the original synthetic embedding.

The training-data scaling experiments further indicate that adaptation performance improves monotonically as additional speakers become available for training. Although reasonable improvements are achieved even with relatively small training sets, larger speaker populations consistently produce better embedding alignment. This observation suggests that additional gains may be obtained by training on larger multi-speaker corpora or more diverse synthetic speech generation

pipelines.

Despite these encouraging results, several limitations remain. First, the current study evaluates adaptation using English speech from the VCTK corpus, and additional experiments on multilingual datasets would provide stronger evidence of language-independent generalization. Second, the present work evaluates adaptation exclusively in the speaker embedding space rather than through downstream applications such as speaker verification, zero-shot text-to-speech, or voice conversion. While improved embedding similarity may benefit these tasks, direct downstream evaluation would provide stronger evidence of practical effectiveness. Finally, the proposed framework is evaluated using a limited number of synthetic speech generation systems. Extending the analysis to additional state-of-the-art neural speech synthesis models would further characterize the generality of the learned adaptation.

Overall, the experimental findings suggest that lightweight embedding-space adaptation offers an effective solution for reducing the domain gap between synthetic and natural speaker embeddings. Unlike approaches that require retraining speaker encoders or modifying downstream speech models, VOCALIS functions as a compact intermediate module that can be readily integrated into existing speech processing pipelines. We hope this work provides a foundation for future research on embedding-space adaptation for synthetic speech and motivates broader investigation into domain-aware speaker representation learning.

VII. CONCLUSION

This paper presented **VOCALIS**, a lightweight embedding-space domain adaptation framework designed to reduce the discrepancy between speaker embeddings extracted from synthetic and natural speech. Rather than modifying pretrained speaker encoders or retraining downstream speech systems, the proposed approach learns a direct nonlinear mapping from synthetic embeddings to the corresponding natural embedding space using paired supervision. Operating entirely in the embedding space enables VOCALIS to remain computationally efficient while maintaining compatibility with existing speaker-conditioned speech processing pipelines.

Comprehensive experiments conducted on the VCTK corpus showed that the proposed method consistently improves the alignment between synthetic and natural speaker embeddings across previously unseen speakers. Compared with both unadapted embeddings and classical linear adaptation methods, VOCALIS achieved the highest embedding similarity while maintaining a compact model architecture. Additional evaluations, including cross-synthesis generalization, multi-seed reproducibility analysis, residual-weight ablation, and training-data scaling experiments, further support the robustness, stability, and data efficiency of the proposed framework. Collectively, these results suggest that explicit embedding-space adaptation provides an effective approach for mitigating the synthetic-to-natural speaker embedding domain gap.

Beyond the specific architecture proposed in this work, the results suggest that embedding adaptation represents a promising research direction for modern speech processing systems that increasingly rely on synthetic speech. Because VOCALIS operates independently of both the speaker encoder and downstream application, the framework could potentially be integrated into a wide range of existing speech technologies, including speaker verification, zero-shot text-to-speech synthesis, voice conversion, and personalized conversational agents. Evaluating these downstream applications directly remains an important direction for future work.

Several opportunities remain for future research. First, the proposed framework should be evaluated on larger and more diverse multilingual speech corpora to assess its generalization across languages and speaking styles. Second, extending the evaluation to additional neural speech synthesis systems would further characterize the robustness of the learned adaptation. Finally, incorporating embedding adaptation directly into end-to-end speech generation or speaker representation learning frameworks represents an interesting direction for jointly optimizing embedding alignment and downstream task performance.

In summary, this work suggests that lightweight embedding-space adaptation can substantially improve the compatibility between synthetic and natural speaker representations while preserving computational efficiency and model independence. We hope that VOCALIS provides a foundation for future research on domain-aware speaker representation learning and encourages broader investigation into robust embedding adaptation for synthetic speech applications.

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